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SWITCHABLE SUPPORTING MECHANISM AND RETICLE CONTAINER APPLYING THE SAME

Abstract

A switchable supporting mechanism adapted for an inner pod base of a dual pod. A switch of an electrostatic discharge (ESD) conduction path is established between the switchable supporting mechanism and the inner pod base. At an instant that a support of the switchable supporting mechanism is pressed downward by a reticle and comes into contact with the inner pod base, the electrostatic discharge conduction path is closed immediately to discharge electrostatic charge accumulated on a front surface of the reticle. In contrast, when the support is not in contact with the inner pod base, the electrostatic discharge conduction path opens. Therefore, the arcing phenomenon due to a voltage difference between electrostatic charge of the reticle and a conventional supporting mechanism can be prevented, and numerous problems such as carbon pollution (C-burst) and damage caused by electrical breakdown of the support can be solved.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/561,362 filed in U.S. on Mar. 5, 2024 the entire contents of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a supporting mechanism of a reticle container, and more particularly to a switchable supporting mechanism and a reticle container applying the same.

DESCRIPTION OF THE PRIOR ART

[0003] A front surface of an extreme ultraviolet (EUV) reticle may carry residual static electricity (for example, positive charge) during exposure of the EUV reticle. An existing reticle support is fixed in an inner pod (EUV inner pod (EIP)) of the reticle container, and is configured to provide electrostatic discharge (ESD). When the inner pod is placed in a scanning machine and is opened, the support, an inner pod base and an apparatus arm are electrically connected to one another and are coupled to ground, hence forming an electrical conduction path. In this case, at an instant that the reticle is placed back into the inner pod base and comes into contact with the support, the residual static electricity on the surface of the reticle is discharged via the electrical conduction path.

[0004] For a reticle having undergone multiple rounds of exposure, charge is continually accumulated so a large amount of charge is present on the surface of the reticle. While such reticle is being placed back into the inner pod base designed with electrostatic discharge, due to an overly large voltage between the reticle and the support as well as the principle of attraction between positive and negative electricity, an arcing phenomenon may easily occur when the reticle approaches the support. As a result, the support may be subject to electrical breakdown, causing related problems such as carbon pollution (C-burst) and damage of the support.

[0005] If the support is in entirety made of an insulating material (without any electrostatic discharge ability), the static electricity accumulated on the reticle cannot be discharged via any path, forming an extremely large potential difference between the reticle and the support. Under the conditions above, in case of any residual substance of a reticle metal coating on the surface of the support, arcing may occur easily to lead to damage much severer than that caused when a support made of an electrostatic discharge material is used. Therefore, there is a need for a solution for solving the problem of arcing and for effectively solving the problem of electrostatic charge on a reticle.

SUMMARY OF THE INVENTION

[0006] In view of the problems above, the present disclosure provides a switchable supporting mechanism adapted for an inner pod base of a dual pod. The switchable supporting mechanism includes: a base, arranged on the inner pod base; a support, having an electrostatic discharge (ESD) material, the support inserting through the base to get close to the inner pod base; and an insulating separator, for separating a contact between the base and the support to cause the support to reciprocate relative to the base, such that the support comes into contact with the inner pod base to establish an electrostatic discharge conduction path, or such that a gap is kept between the support

and the inner pod base to disconnect the electrostatic discharge conduction path.

[0007] In a specific embodiment, when the support does not receive any external force, the support departs from the inner pod base such that the gap is kept between the support and the inner pod base to disconnect the electrostatic discharge conduction path; when the support receives an external force, the support comes into contact with the inner pod base to establish the electrostatic discharge conduction path.

[0008] In a specific embodiment, the base includes a mounting portion and a mounting hole penetrating the mounting portion and the base, and the insulating separator includes an abutting portion and a connecting portion. The abutting portion abuts against an upper surface of the mounting portion, and the connecting portion fixes the insulating separator to the mounting portion via the mounting hole.

[0009] In a specific embodiment, the support includes a carrying portion, a contact portion, a pin portion located between the carrying portion and the contact portion, and the insulating separator has a through hole in communication with the mounting hole. The pin portion is inserted into the through hole and the mounting hole, such that the contact portion is adjacent to the inner pod base and the carrying portion is located on an upper surface of the abutting portion.

[0010] In a specific embodiment, the insulating separator includes at least one mounting structure located outside the connecting portion. The mounting structure passes through an extension expansion hole of the mounting hole and is fixed on a lower surface of the mounting portion.

[0011] In a specific embodiment, the upper surface of the mounting portion is provided with a mounting groove, the connecting portion of the insulating separator is arranged in the mounting groove, and the abutting portion abuts against an upper surface of a periphery of the mounting groove such that the insulating separator is fixed in the mounting portion.

[0012] In a specific embodiment, the insulating separator is made of an elastic material, such that when an external force is applied to or released from the support, the insulating separator is accordingly elastically compressed and deformed or elastically released and reset to cause the support to reciprocate.

[0013] In a specific embodiment, an insulating anti-detachment element is further included. The insulating anti-detachment element is arranged between the base and the support to prevent the support from departing from between the base and the insulating separator.

[0014] In a specific embodiment, the support includes a carrying portion, a contact portion, a pin portion located between the carrying portion and the contact portion, and the insulating anti-detachment element sleeves the pin portion. In a specific embodiment, the base includes a mounting portion and a mounting hole penetrating the mounting portion and the base, and an upper surface of the mounting portion is provided with a mounting groove. The support includes a carrying portion, a contact portion, and a pin portion located between the carrying portion and the contact portion. The insulating separator is arranged in the mounting groove and has a through hole in communication with the mounting hole, and the pin portion is inserted into the through hole and the mounting hole, such that the contact portion is adjacent to the inner pod base and a lower surface of the carrying portion is located on an upper surface of the insulating separator.

[0015] In a specific embodiment, an electrostatic discharge element is further arranged at a top and/or a bottom of the support.

[0016] In a specific embodiment, the support and the inner pod base establish the electrostatic discharge conduction path, which is connected to an apparatus arm to release electrostatic charge to the ground via the apparatus arm.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] Reference can be made to the drawings and description below to better understand the present disclosure. Non-limiting and non-exhaustive embodiments are described with reference to the drawings below. It should be noted that the components in the drawings are not necessarily drawn according to their actual sizes, and are depicted to focus on the description of structures and principles.

[0018] FIG. 1 is a perspective diagram provided with an inner pod base according to an embodiment of the present disclosure.

[0019] FIG. 2 is a partial enlarged diagram of area A in FIG. 1.

[0020] FIG. 3 is an exploded diagram of a supporting mechanism according to a first embodiment of the present disclosure.

[0021] FIG. 4A is a cross-sectional schematic diagram of the supporting mechanism according to the first embodiment of the present disclosure taken along line aa in FIG. 2 when an electrostatic discharge conduction path is not formed.

[0022] FIG. 4B is a cross-sectional schematic diagram of the supporting mechanism according to the first embodiment of the present disclosure taken along the line aa in FIG. 2 when an electrostatic discharge conduction path is formed.

[0023] FIG. 5 is an exploded diagram of a supporting mechanism according to a second embodiment of the present disclosure.

[0024] FIG. 6A is a cross-sectional schematic diagram of the supporting mechanism according to the second embodiment of the present disclosure when an electrostatic discharge conduction path is not formed.

[0025] FIG. 6B is a cross-sectional schematic diagram of the supporting mechanism according to the second embodiment of the present disclosure when an electrostatic discharge conduction path is formed.

[0026] FIG. 7 is an exploded diagram of a supporting mechanism according to a third embodiment of the present disclosure.

[0027] FIG. 8A is a cross-sectional schematic diagram of the supporting mechanism according to the third embodiment of the present disclosure when an electrostatic discharge conduction path is not formed.

[0028] FIG. 8B is a cross-sectional schematic diagram of the supporting mechanism according to the third embodiment of the present disclosure when an electrostatic discharge conduction path is formed.

[0029] FIG. 9A is a cross-sectional schematic diagram of the supporting mechanism of a first variation example of the present disclosure when an electrostatic discharge conduction path is not formed.

[0030] FIG. 9B is a cross-sectional schematic diagram of the supporting mechanism of the first variation example of the present disclosure when an electrostatic discharge conduction path is formed.

[0031] FIG. 10A is a cross-sectional schematic diagram of the supporting mechanism of a second variation example of the present disclosure when an electrostatic discharge conduction path is not formed.

[0032] FIG. 10B is a cross-sectional schematic diagram of the supporting mechanism of the second variation example of the present disclosure when an electrostatic discharge conduction path is formed.

[0033] FIG. 11A is a cross-sectional schematic diagram of the supporting mechanism of a third variation example of the present disclosure when an electrostatic discharge conduction path is not formed.

[0034] FIG. 11B is a cross-sectional schematic diagram of the supporting mechanism of the third

variation example of the present disclosure when an electrostatic discharge conduction path is formed.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0035] Referring to FIG. 1, the present disclosure provides a reticle container having a reticle supporting mechanism with a switchable electrostatic discharge (ESD) path. The reticle container is a dual pod, and includes an outer pod and an inner pod accommodated in the outer pod. The reticle supporting mechanism of the present disclosure is adapted for a base of the inner pod. In the embodiment in FIG. 1, an inner pod base **100** included in the inner pod is shown while a cover of the inner pod is omitted. The inner pod base **100** provides multiple supporting mechanisms **20** for carrying a reticle R.

[0036] Referring to FIG. 2, four corners of the reticle R respectively correspond to the four supporting mechanisms **20**. The inner pod base **100** of this embodiment has a groove in which the supporting mechanisms **20** are arranged. The present disclosure does not define that the inner pod base **100** necessarily has a groove, and regions for placing the reticle and arranging the supporting mechanisms **20** may be coplanar. The supporting mechanism **20** includes a base **21**, an insulating separator **22** and a support **23**. The base **21** is arranged at the inner pod base **100** of the reticle container, the insulating separator **22** is arranged on the base **21**, and the support **23** is arranged adjacent to the inner pod base **100** via the insulating separator **22** and functions as a switch of an electrostatic discharge path. The reticle R is placed on the support **23**. The insulating separator **22** is located between the support **23** and the base **21**, and is for separating a contact between the base **21** and the support **23** to maintain electrical disconnection between the base **21** and the support **23**. The insulating separator **22** determines whether the support **23** and the inner pod base **100** are to come into contact with each other. In a non-deformed state of the insulating separator **22**, the support **23** and the inner pod base **100** are not in direct contact with each other. In a deformed state of the insulating separator **22**, the support **23** and the inner pod base **100** are in direct contact with each other, with associated details described below. Refer to FIG. 3 showing an exploded diagram of the supporting mechanism **20** according to a first embodiment of the present disclosure. The base **21** includes a seat **215**, a limiting portion **212** and a mounting portion **211**. This embodiment includes two limiting portions **212** respectively extending upward from two ends of the seat **215**. The mounting portion **211** bulges upwards from the seat **215**. The mounting portion **211** has a mounting hole **213** at a center thereof, wherein the mounting hole **213** penetrates the seat **215** and the mounting portion **211** such that the insulating separator **22** is arranged in the mounting hole **213**.

[0037] The insulating separator **22** is substantially made of an electrically insulative and elastic material. When an external force is applied to or released from the support **23**, the insulating separator **22** is accordingly elastically compressed and deformed or elastically released and reset to cause the support **23** to reciprocate. The insulating separator **22** includes an abutting portion **221** and a connecting portion **222**. The connecting portion **222** can couple the insulating separator **22** to the seat **21** via the mounting hole **213**, and the abutting portion **221** abuts against an upper surface of the mounting portion **211** at this point in time. Moreover, the abutting portion **221** is configured between the mounting portion **211** and a portion of the support **23**. The insulating separator **22** further includes a through hole **225**. The through hole **225** penetrates upper and lower surfaces of the insulating separator **22**, and corresponds to a position of the mounting hole **213** to communicate with each other. A portion of the support **23** is inserted into the through hole **225** and the mounting hole **213**. In this embodiment, the insulating separator **22** further includes a mounting structure **224** protruding outward from a surface of the connecting portion **222**. The mounting hole **213** has an extension expansion hole **214** corresponding to the mounting structure **224**, for the mounting structure **224** to pass through and for the insulating separator **22** to be mounted in the mounting hole **213**. Preferably, a lower edge of the insulating separator **22** further includes an engaging portion **223** for engaging the support **23** with the insulating separator **22**, so as to prevent the

support **23** from departing upward.

[0038] The support **23** has an electrostatic discharge material. The support **23** includes a carrying portion **231**, a contact portion **234**, and a pin portion **232** arranged between the carrying portion **231** and the contact portion **234**. The carrying portion **231** is for carrying the reticle **R** and may have an arc or a pattern, such that the carrying portion **231** and the reticle **R** are in point contact to reduce the contact area of the reticle **R** and reduce dust generation. The pin portion **232** passes through the through hole **225** of the insulating separator **22** to have the support **23** being adjacent to the insulating separator **22**. The contact portion **234** of the support **23** is then placed into the mounting hole **213** to a position near the inner pod base **100**. The carrying portion **231** is located on an upper surface of the abutting portion **221**.

[0039] The supporting mechanism **20** further includes an insulating anti-detachment element **233** arranged between the seat **21** and the support **23**. In this embodiment, the insulating anti-detachment element **233** is arranged at the pin portion **232** of the support **23**, and the insulating anti-detachment element **233** and the engaging portion **223** of the insulating separator **22** may be engaged with each other to prevent the support **23** from departing from the insulating separator **22**. Moreover, since the insulating separator **22** is fixed on the seat **21** via the abutting portion **221** and the mounting structure **224**, the support **23** can be prevented from departing from the seat **21** and the insulating separator **22** by means of interlocking connection. The insulating anti-detachment element **233** is adapted for any form of the insulating separator, and can be matched and combined as desired and is not limited to the form disclosed in this embodiment.

[0040] In continuation from the description of the supporting mechanism **20** above, when the supporting mechanism **20** is applied to the inner pod base **100**, the seat **21** may be fixed on the inner pod base **100** by any conventional means. Next, the mounting structure **224** of the insulating separator **22** is aligned with and placed into the extension expansion hole **214** of the mounting hole **213** of the seat **21**, and the insulating separator **22** is rotated to mismatch the mounting structure **224** and the extension expansion hole **214**, such that the insulating separator **22** is fixed to the seat **21**. Lastly, the support **23** is placed into the through hole **225** of the insulating separator **22**, and an external force is slightly applied to have the insulating anti-detachment element **233** pass through a lower edge of the insulating separator **22** and be engaged with the engaging portion **223**. At this point in time, the contact portion **234** of the support **23** protrudes outside the mounting hole **214** of the seat **21**, and is adjacent to the inner pod base **100**. The mounting sequences include but are not limited to the steps above, and the mounting sequences of the elements can be adjusted according to requirements.

[0041] In this embodiment, the support **23** and the seat **21** have an electrostatic discharge function, and can be, for example, made of an electrostatic discharge material or have surfaces coated with an electrostatic discharge coating material; the insulating separator **22** is made of an insulating material or has a surface coated with an insulating coating material. Preferably, surface resistivities of the support **23** and the seat **21** range between $10.\sup{.4} \Omega/\text{sq}$ and $10.\sup{.10} \Omega/\text{sq}$, and a surface resistivity of the insulating separator **22** is equal to or greater than $10.\sup{.10} \Omega/\text{sq}$. In this embodiment, the material forming the insulating separator **22** may include polyethylene (PE), polypropylene (PP), polyethylene terephthalate (PET), thermoplastic elastomer (TPR), thermoplastic polyester elastomer (TPEE), fluorinated rubber (FKM), or any combination of the above. The materials forming the support **23** and the seat **21** may include polyetheretherketone (PEEK), polyimide (PI), or any combination of the above.

[0042] The insulating separator **22** has an elastic structure. When a downward pressure (or referred to as an external force) generated by placing a reticle is applied to the support **23**, the insulating separator **22** receives the abutment between the seat **21** and the support **23** and produces elastic structural deformation such as elastic compression deformation, such that the support **23** can reciprocate and move downward relative to the seat **21** until the contact portion **234** comes into contact with the inner pod base **100** so as to establish an electrostatic discharge conduction path.

The electrostatic charge on the reticle can be electrostatically discharged via the support **23** to the inner pod base **100**. When the inner pod base **100** is located within a scanning machine, the scanning machine is provided with an apparatus arm and is grounded. Once the inner pod base **100** comes into contact with the apparatus arm, it may form one of the electrostatic discharge conduction paths via the apparatus arm and the ground and thus has an electrostatic discharge function.

[0043] When a reticle is removed from the support **23** or when no reticle is placed on the support **23**, the insulating separator **22** is not subject to the abutment between the seat **21** and the support **23**, and is elastically released at this point in time to restore its original elastic height, such that a gap *d* is kept between the contact portion **234** of the support **23** and the inner pod base **100** to disconnect the electrostatic discharge conduction path. In a specific embodiment, by separating the contact between the seat **21** and the support **23** by the insulating separator **22** and according to whether the support **23** receives an external force, it is determined whether the support **23** and the inner pod base **100** are to come into contact with each other, accordingly achieving the technical effect and object of the switch for establishing an electrostatic discharge conduction path between the supporting mechanism **20** and the inner pod base **100**.

[0044] In continuation of the description above, referring to FIG. 4A, after exposure, a front surface of the reticle R carries residual electrostatic charge, for example, positive charge. When the support **23** does not receive any external force of a downward pressure from the reticle R, the gap *d* is present between the support **23** and the inner pod base **100**. At this point in time, the support **23** lacks an ability of attracting the electrostatic charge on the front surface of the reticle R, and thus does not conduct any electricity through the ground GND. In other words, the support **23** cannot attract the positive electricity of the reticle R from a ground terminal to the ground GND, and thus the electrostatic discharge conduction path is disconnected at this point in time.

[0045] Referring to FIG. 4B, when the reticle R is placed at the support **23**, the support **23** receives an external force and is pushed downward to force the insulating separator **22** to become elastically deformed until surfaces of the contact portion **234** of the support **23** and the inner pod base **100** are in contact with each other. At this point in time, the electrostatic discharge conduction path is formed. At this point in time, the support **23** has an ability of attracting the electrostatic charge on the front surface of the reticle R, and an electrostatic discharge conduction path along the support **23**, the inner pod base **100** and the ground GND is established to discharge electrostatic charge of the reticle R.

[0046] By establishing the switch of an electrostatic discharge conduction path, the object of the present disclosure is to solve the numerous problems of an arcing phenomenon due to a voltage difference when a reticle with a large amount of static electricity accumulated on a surface thereof approaches a supporting mechanism, as well as carbon pollution (C-burst) and damage caused by electrical breakdown of the supporting mechanism. Moreover, an effect of quickly electrostatically discharging the residual static electricity on the surface of a reticle after exposure can be achieved.

[0047] FIG. 5 shows a supporting mechanism **30** according to a second embodiment of the present disclosure. In the supporting mechanism **30**, the elements denoted by the same numerals as the supporting mechanism **20** of the first embodiment have the same functions, and only structural differences are described below. The supporting mechanism **30** includes the seat **21**, the insulating separator **22**, an insulating anti-detachment element **34** and the support **23**. A groove **314** is provided annularly on an upper surface of the mounting hole **213**. The through hole **225** penetrates the upper and lower surfaces of the insulating separator **22**, and corresponds to the position of the mounting hole **213** to communicate with each other. A portion of the support **23** is inserted in the through hole **225** and the mounting hole **213**.

[0048] A top of the insulating separator **22** has the abutting portion **221** made of an electrically insulative and elastic material, and causes the support **23** to reciprocate relative to the seat **21** when an external force is applied to or released from the support **23**. The connecting portion **222** of the

insulating separator **22** corresponds in shape to the groove **314**, and the two can be mutually installed and fixed so as to couple the insulating separator **22** and the seat **21** with each other. In this embodiment, the insulating anti-detachment element **34** is an annular member made of an elastic material, and the pin portion **232** of the support **23** has a groove **333** around a surface thereof, such that the insulating anti-detachment element **34** sleeves the groove **333**.

[0049] Referring to both FIG. **6A** and FIG. **6B**, a seat **31** is provided with a limiting structure **316** located inside the mounting hole **213** to abut against the insulating anti-detachment element **34**. Once the pin portion **232** of the support **23** is inserted into the through hole **225** of the insulating separator **22** and the mounting hole **213** of the seat **31**, the limiting structure **316** abuts against the insulating anti-detachment element **34** to prevent disengagement of the support **23**. Since the support **23** functions as a switch of an electrostatic discharge path, when an external force is applied to or released from the support **23**, the insulating separator **22** is accordingly elastically compressed and deformed or elastically released and reset to cause the support **23** to reciprocate. As shown in FIG. **6A**, the insulating separator **22** is elastically released to restore its original elastic height, such that the gap **d** is kept between the contact portion **234** of the support **23** and the inner pod base **100** to open the electrostatic discharge conduction path. To prevent the support **23** from moving upward and departing from one or both of the base **31** and the insulating separator **22**, the limiting structure **316** is used to abut against and restrict the insulating anti-detachment element **34** and to keep the insulating anti-detachment element **34** in the mounting hole **213**, further preventing the support **23** from moving excessively upward and hence preventing the pin portion **232** from falling out of the mounting hole **213**.

[0050] As shown in FIG. **6B**, the insulating separator **22** is configured to allow the support **23** to move downward relative to the base **21** until the contact portion **234** comes into contact with the inner pod base **100**, so as to establish the electrostatic discharge conduction path. At this point in time, the limiting structure **316** does not abut against the insulating anti-detachment element **34**. Moreover, because the insulating anti-detachment element **34** sleeves the groove **333** such that it moves downward along with the support **23**, the smoothness of reciprocation of the support **23** is unaffected.

[0051] FIG. **7** shows a supporting mechanism **40** according to a third embodiment of the present disclosure. In the supporting mechanism **40**, the elements denoted by the same numerals as the supporting mechanism **20** of the first embodiment and the supporting mechanism **30** of the second embodiment have the same functions, and only structural differences are described below. The insulating separator **22** has an electrically insulative and elastic material, and is an annular member. An upper surface of the insulating separator **22** is defined as the abutting portion **221**, and the connecting portion **222** is defined between the upper surface and the lower surface. The connecting portion **222** of the insulating separator **22** corresponds in shape to a groove **414**, and the two can be mutually installed and fixed so as to couple the insulating separator **22** and the seat **21** with each other. The abutting portion **221** is configured between the mounting portion **211** and the support **23**. The insulating anti-detachment element **42** is an annular member made of an elastic material, and the pin portion **232** of the support **23** has a groove **433** around a surface thereof, and the insulating anti-detachment element **42** is for sleeving the groove **433**. The support **23** is placed into an annular hollow of the insulating separator **22** in a direction toward the seat **21**, and a slight external force is applied to push the insulating anti-detachment element **42** to below a limiting structure **416**, such that the support **23** and the seat **21** become coupled with each other via the insulating separator **22**. When an external force is applied to or released from the support **23**, the elastic deformation of the insulating separator **22** causes the support **23** to reciprocate relative to the seat **21**.

[0052] Referring to FIG. **8A**, when the reticle **R** is not yet placed on the support **23**, because the insulating separator **22** is not yet compressed or deformed and is maintained with its original shape and height, the insulating anti-detachment element **42** abuts against below the limiting structure **416**, such that the support **23** cannot move upward and the support **23** is not in contact with the

inner pod base **100** at this point in time. The gap is present between the contact portion **234** of the support **23** and the surface of the inner pod base **100**. At this point in time, the electrostatic discharge conduction path between the support **23** and the inner pod base **100** is disconnected. When the electrostatic discharge conduction path of the supporting mechanism **40** is disconnected, despite that residual static electricity exists on the surface of the reticle R after exposure, the arcing phenomenon does not occur if the reticle R approaches the support **23** at this point in time, and pollution caused by arcing can be prevented.

[0053] Referring to FIG. **8B**, when the reticle R is placed at the support **23**, the support **23** receives an external force of a downward pressure by the reticle R and is pushed downward to force the insulating separator **22** to become compressed or deformed to allow the support **23** to move downward relative to the base **21**, and the insulating anti-detachment element **42** also moves downward along with the support **23** until surfaces of the contact portion **234** of the support **23** and the inner pod base **100** are in contact with each other. At this point in time, the electrostatic discharge conduction path between the bottom of the support **23** and the inner pod base **100** is formed. When the electrostatic discharge conduction path of the supporting mechanism **40** is formed, the positive charge on the surface of the reticle R is discharged through a path along the support **23**, the inner pod base **100**, the apparatus arm and the ground (not shown).

[0054] In a variation example of the present disclosure, in addition to being in entirety made of an electrostatic discharge material, the support **23** can also be partially made of electrostatic discharge material, for example, the top and/or the bottom of the support **23** is further provided with an electrostatic discharge element. FIG. **9A** and FIG. **9B** show a first variation example of a supporting mechanism of the present disclosure. The support **23** further includes an electrostatic discharge element **24** arranged at the top of the support **23**. More specifically, the electrostatic discharge element **24** is arranged at the carrying portion **231** of the support **23** and is detachably connected to the support **23**.

[0055] As shown in FIG. **9B**, when the reticle R is placed on the support **23**, the reticle R comes into contact with the electrostatic discharge element **24**, and then the electrostatic discharge element **24** and the support **23** move downward together until the contact portion **234** of the support **23** comes into contact with the inner pod base **100**. At this point in time, the positive charge on the surface of the reticle R is discharged through a path along the electrostatic discharge element **24**, the support **23**, the inner pod base **100**, the apparatus arm and the ground (not shown).

[0056] FIG. **10A** and FIG. **10B** show a supporting mechanism of a second variation example. The support **23** further includes an electrostatic discharge element **25** arranged at the bottom of the support **23**; for example, the electrostatic discharge element **25** is provided below the contact portion **234**, or the electrostatic discharge element **25** is provided on the pin portion **232** and the contact portion **234**. More specifically, the electrostatic discharge element **25** is arranged at the support **23** and is detachably connected to the support **23**.

[0057] As shown in FIG. **10B**, when the reticle R is placed on the support **23**, the reticle R first comes into contact with the support **23** and moves downward together with the support **23**, such that the electrostatic discharge element **25** at the bottom of the support **23** comes into contact with the inner pod base **100**. At this point in time, the positive charge on the surface of the reticle R is discharged through a path along the support **23**, the electrostatic discharge element **25**, the inner pod base **100**, the apparatus arm and the ground (not shown).

[0058] FIG. **11A** and FIG. **11B** show a supporting mechanism of a third variation example. The support **23** further includes a first electrostatic discharge element **24** and a second electrostatic discharge element **25** respectively arranged at the top and the bottom of the support **23**. More specifically, the first electrostatic discharge element **24** is arranged at the carrying portion **231** of the support **23** and is detachably connected to the support **23**. The second electrostatic discharge element **25** is arranged at the contact portion **234** of the support **23** and is detachably connected to the support **23**.

[0059] As shown in FIG. 11B, when the reticle R is placed on the support 23, the reticle R comes into contact with the first electrostatic discharge element 24, and then the first electrostatic discharge element 24 and the support 23 move downward together, until the second electrostatic discharge element 25 at the bottom of the support 23 comes into contact with the inner pod base 100. At this point in time, the positive charge on the surface of the reticle R is discharged through a path along the first electrostatic discharge element 24, the support 23, the second electrostatic discharge element 25, the inner pod base 100, the apparatus arm and the ground (not shown).

[0060] By additionally arranging the electrostatic discharge elements 24 and 25 of the first to third variation examples at the support 23, the electrostatic discharge elements 24 and 25 can be easily replaced, so as to prevent wear caused by the contact between the reticle R and the carrying portion 231 of the support 23 and/or wear caused by the contact between the bottom of the support 23 and the inner pod base 100 from affecting the effect of the electrical conduction path. Regardless of how the structural design of the support 23 is modified, any support 23 functioning as a switch of an electrostatic discharge path is encompassed within the scope of the present disclosure.

Claims

1. A switchable supporting mechanism adapted for an inner pod base of a dual pod, the switchable supporting mechanism comprising: a base, arranged on the inner pod base; a support, having an electrostatic discharge material, the support inserting through the base to get close to the inner pod base; and an insulating separator, selectively separating a contact between the base and the support to cause the support to reciprocate relative to the base, whereby causing the support able to come into contact with the inner pod base to establish an electrostatic discharge conduction path, or keeping a gap between the support and the inner pod base to disconnect the electrostatic discharge conduction path.
2. The switchable supporting mechanism according to claim 1, wherein when the support does not receive any external force, the support departs from the inner pod base such that the gap is kept between the support and the inner pod base to disconnect the electrostatic discharge conduction path; when the support receives an external force, the support comes into contact with the inner pod base to establish the electrostatic discharge conduction path.
3. The switchable supporting mechanism according to claim 1, wherein the base comprises a mounting portion and a mounting hole penetrating the mounting portion and the base, and the insulating separator comprises an abutting portion and a connecting portion; the abutting portion abuts against an upper surface of the mounting portion, and the connecting portion fixes the insulating separator to the mounting portion via the mounting hole.
4. The switchable supporting mechanism according to claim 3, wherein the support comprises a carrying portion, a contact portion, a pin portion located between the carrying portion and the contact portion, the insulating separator has a through hole in communication with the mounting hole, and the pin portion is inserted into the through hole and the mounting hole, such that the contact portion is adjacent to the inner pod base and the carrying portion is located on an upper surface of the abutting portion.
5. The switchable supporting mechanism according to claim 3, wherein the insulating separator comprises at least one mounting structure located outside the connecting portion, and the mounting structure passes through an extension expansion hole of the mounting hole and is fixed on a lower surface of the mounting portion.
6. The switchable supporting mechanism according to claim 3, wherein the upper surface of the mounting portion is provided with a mounting groove, the connecting portion of the insulating separator is arranged in the mounting groove, and the abutting portion abuts against an upper surface of a periphery of the mounting groove, such that the insulating separator is fixed in the mounting portion.

- 7.** The switchable supporting mechanism according to claim 1, wherein the insulating separator is made of an elastic material, and when an external force is applied to or released from the support, the insulating separator is accordingly elastically compressed and deformed or elastically released and reset to cause the support to reciprocate.
- 8.** The switchable supporting mechanism according to claim 1, further comprising: an insulating anti-detachment element, arranged between the base and the support to prevent the support from departing from between the base and the insulating separator.
- 9.** The switchable supporting mechanism according to claim 8, wherein the support comprises a carrying portion, a contact portion, a pin portion located between the carrying portion and the contact portion, and the insulating anti-detachment element sleeves the pin portion.
- 10.** The switchable supporting mechanism according to claim 1, wherein the base comprises a mounting portion and a mounting hole penetrating the mounting portion and the base, and a mounting groove is provided on an upper surface of the mounting portion; the support comprises a carrying portion, a contact portion, a pin portion located between the carrying portion and the contact portion; and the insulating separator is arranged in the mounting groove and has a through hole in communication with the mounting hole, and the pin portion is inserted into the through hole and the mounting hole, such that the contact portion is adjacent to the inner pod base and a lower surface of the carrying portion is located on an upper surface of the insulating separator.
- 11.** The switchable supporting mechanism according to claim 1, wherein a top and/or a bottom of the support is further arranged with an electrostatic discharge element.
- 12.** The switchable supporting mechanism according to claim 1, wherein the support and the inner pod base establish the electrostatic discharge conduction path, which is connected to an apparatus arm to release electrostatic charge to the ground via the apparatus arm.
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